C.R.E.D.I.T.

PROTOCOL STATUS: OPERATIONAL

Carbon & Regenerative Ecological Derivatives Investment Token

Financing the future of the planet — one acre at a time.
Bridging the gap between smallholder farmers and institutional capital via the Transparent Ledger.

01. The Market Failure

The global agricultural and climate finance systems are broken. C.R.E.D.I.T. addresses three interconnected failures locking farmers out of capital markets.

ISSUE ONE

The $170B Gap

50-100%

APR LENDING

Farmers are forced to borrow from local moneylenders, trapping generations in debt and preventing investment in regenerative land practices.

ISSUE TWO

Climate Exclusion

< 1%

CLIMATE FINANCE

Smallholder farmers have the land and incentive to adopt climate-positive practices, but verification costs lock them entirely out of Voluntary Carbon Markets.

ISSUE THREE

The Trust Deficit

NULL

VERIFICATION

Corporate ESG buyers face a greenwashing crisis. They cannot confidently verify if the credits they buy represent actual carbon reductions.

02. Market Opportunity

By solving verification at scale, C.R.E.D.I.T. unlocks both the supply of smallholder ecological assets and the demand for auditable environmental exposure.

DEMAND

Carbon Market Expansion

$50B+

VCM SCALE BY 2030

Compliance pressure and corporate net-zero commitments are increasing demand for higher-quality, auditable environmental assets.

SUPPLY

Massive Origin Base

500M+

SMALLHOLDER FARMS

Smallholder farms represent one of the largest untapped real-world asset bases for both regenerative value creation and harvest-linked finance.

FOCUS

India & Southeast Asia

HIGH

IMPACT DENSITY

These regions sit at the intersection of farmer debt stress, regenerative potential, and demand for better climate-finance rails.

03. Why Now?

C.R.E.D.I.T. becomes possible only because three shifts have converged: better sensing, better cryptographic infrastructure, and stronger demand for provable environmental assets.

TECHNOLOGY

Democratized dMRV APIs

Multispectral and SAR satellite imagery are now affordable and frequent enough to support rural auditing at protocol scale.

CRYPTOGRAPHY

Maturing ZK Infrastructure

Zero-knowledge systems make it increasingly realistic to verify outcomes without exposing sensitive farmer data publicly.

REGULATION

Harder ESG Scrutiny

Buyers and regulators increasingly require auditable proof, making opaque legacy offsets structurally weaker over time.

04. Ecosystem at a Glance

An end-to-end marketplace routing institutional capital down to grassroots smallholders, secured entirely by mathematical verification and automated smart contracts.

1

REAL WORLD

Smallholder Farmers implement regenerative practices monitored continuously by IoT Sensors & Satellite Data.

→

2

CREDIT PROTOCOL

The CREDIT-Guard Oracle Network generates ZK-verified proofs, triggers Smart Contracts, and archives raw data on BNB Greenfield.

→

3

THE MARKET

CREDIT-VCC & ACFC tokens are minted to DEX marketplaces where Institutional Investors provide liquidity seamlessly.

05. Verified Carbon Credits (CREDIT-VCC)

An end-to-end data integrity standard using dMRV to mint tokens directly tied to scientifically grounded, verifiable real-world outcomes.

Zero-Trust Issuance

- **Ingestion:** Multispectral satellite & IoT sensors continuously stream ground data.
- **Verification:** ZK-proofs generated via the CREDIT-Guard Oracle Network.
- **Archival:** Raw data permanently archived via BNB Greenfield decentralised storage.
- **Minting:** Smart contract deterministically issues ERC-1155 token to the farmer.

1 TOKEN = 1 TONNE CO2E

DATA CONTINUITY MODEL

1 **Physical Actions** Farmer scales zero-tillage & cover planting.

2 **Oracle ZK Proof** Data processed. Mathematical proof encoded without metadata exposure.

3 **DEX Execution** Institutional capital acquires instantly verifiable ERC-1155 assets.

06. The Verification-to-Minting Pipeline

The full lifecycle of a CREDIT token from a real-world agricultural event to a tradeable on-chain asset.

STAGE 1

Data Ingestion

IoT devices and multispectral/SAR satellite APIs continuously stream environmental data. Every incoming data packet is timestamped and cryptographically signed.

STAGE 2

ZK Verification

The oracle network processes incoming data against the Regen-Standard criteria. A Zero-Knowledge proof is generated confirming pass/fail without exposing farmer privacy.

STAGE 3

Greenfield Archival

Before minting, the complete raw data payload is uploaded to BNB Greenfield. The storage object ID is returned to provide an auditable public foundation.

STAGE 4

On-Chain Minting

The smart contract receives the ZK proof, verifies its validity on-chain, confirms the Greenfield reference, and mints the token to the farmer's wallet deterministically.

07. Agri-Commodity Forward Contracts (CREDIT-ACFC)

Solving the working capital problem instantly by bringing legally enforceable forward contracts on-chain, uniquely tailored for unbanked smallholders.

Defeating Predatory APRs

Farmers declare a percentage of expected future harvest (e.g., 4 tonnes of rice) minting unique **ERC-721 tokens** representing defined delivery terms at a discounted spot rate.

Investors provide immediate capital via DEX marketplace, securing the protocol's liquidity. Settlements happen via smart contract upon oracle data verification of the yield.

LIQUIDITY	YIELD SETTLEMENT
Immediate	Programmatic

CREDIT-ACFC SPECS

ASSET STANDARD
ERC-721 NFT

UNDERLYING
Physical Ag-Yield

INSURANCE
Treasury Buffered

08. The Institutional Moat

Most carbon systems stop at registry UI or marketplace access. C.R.E.D.I.T. integrates verification, provenance, issuance, and settlement into one trust architecture.

DISINTERMEDIATION

Protocol, Not Manual Brokerage

Asset creation is tied to oracle-triggered rules rather than expensive broker-led workflows and fragmented off-chain approval chains.

PROVENANCE

Immutable Audit Base

Greenfield-backed data continuity gives buyers a tamper-evident evidence trail instead of a black-box registry assertion.

PRIVACY

Proof Without Data Leakage

The protocol is designed to make ecological verification auditable without exposing exact farm coordinates or sensitive production information.

09. The Protocol Architecture

A triple-layered technology stack designed for decentralized security, high resilience, and unalterable verification.

LAYER 1 · SETTLEMENT

BNB Chain

Provides EVM compatibility, low transaction fees, and high throughput. Handles atomic settlements, DEX listings, and minting of ERC-1155 & ERC-721 tokens effectively.

LAYER 2 · STORAGE

BNB Greenfield

Native decentralized storage network hosting all multi-gigabyte dMRV data (Raw satellite tiles, metadata, documents) permanently linked via on-chain asset metadata.

LAYER 3 · ORACLE

CREDIT-Guard

Custom oracle network validating ZK Proofs representing environmental pass/fail criteria. Blocks bad data ingestion without compromising proprietary farmland coordinates.

10. Business Model

Revenue starts with protocol fees on issuance, trading, and settlement. The current marketplace implementation already enforces a 2.5% fee.

TODAY

Protocol Marketplace Fees

2.5%

CURRENT FEE

Applied to market activity across the existing marketplace flow as the first monetization layer of the protocol.

EXPANSION

Issuance + Settlement

As verification matures, the protocol can capture value at the points where new assets are created, financed, and settled.

THESIS

Own the Trust Rail

The larger opportunity is becoming the issuance and trust layer for ecological and agricultural RWAs that currently cannot reach capital markets credibly.

11. Native Tokenomics ($CREDIT)

Driving protocol alignment across institutional buyers, farmers, and decentralised node operators without relying on runaway speculation.

VALUE ACCRUAL

Fee Governance

Every commercial action across the network (Carbon purchase, ACFC settlement) enforces a protocol marketplace fee.

- **25% Discount** when fees are paid using $CREDIT, creating perpetual demand vectors.
- **75% Yield** directed toward the DAO for operational development and continuous R&D.

SYSTEMIC PROTECTION

Climate Insurance Treasury

Agricultural income is highly exposed to catastrophic climate-scale events (flash floods, droughts, pests).

- **25% of all Protocol Fees** are automatically funnelled to the community risk backstop.
- **Programmatic Payouts:** In disaster events verified by the Oracle, the Treasury covers investor shorts and shields the farmer instantly automatically.

12. Protocol Roadmap

The path from prototype to production is straightforward: deepen verification, harden underwriting, and onboard real-world distribution partners.

PHASE 1 · ACTIVE

Testnet Deployment

Core contracts, oracle-driven payload generation, Greenfield-linked metadata flow, and marketplace demo are already in place at MVP level.

PHASE 2 · NEXT

Mainnet + Underwriting

Strengthen verification quality, add more robust risk controls for forward contracts, and move toward production-grade settlement logic.

PHASE 3 · SCALE

India & SEA Rollout

Expand through farmer cooperatives, auditors, and buyer channels to turn regenerative agriculture into a repeatable RWA origination pipeline.

13. Team

C.R.E.D.I.T. is being built as a cross-functional protocol effort spanning smart contracts, oracle/data systems, and product execution. This slide is role-based because the repo does not include public founder bios.

ROLE 1

Protocol & Product

Owns market thesis, product scope, RWA design, and how farmer financing and buyer trust connect in one protocol narrative.

ROLE 2

Smart Contract Engineering

Builds ERC-1155 carbon credit issuance, ERC-721 forward contracts, and marketplace logic on BNB infrastructure.

ROLE 3

Oracle, Data & Frontend

Builds the data-to-mint pipeline, Greenfield-linked provenance flow, and the product interfaces used to demonstrate issuance and market access.

C.R.E.D.I.T.

JOIN THE REGENERATIVE ECONOMY

The Future Is Verified.

C.R.E.D.I.T. turns regenerative agriculture into an investable RWA category by making ecological performance verifiable, future yield financeable, and provenance visible to the market.